Building a Hybrid Search System for Academic Documents using Elasticsearch

Course	CO5135 - Big Data
Group	11
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Project Topic

Building a Hybrid Search System for Academic Documents using Elasticsearch

Context

Efficient academic document search is essential for students and researchers. Traditional search systems reveal many limitations when handling massive volumes of data.

Objective

Apply a Hybrid Search architecture to deliver the most accurate results, positioning Elasticsearch as the core engine search to handle both text and vector data seamlessly

Core Technology

Elasticsearch (8.x series). We leverage its modern "Retrievers" API and built-in Reciprocal Rank Fusion (RRF) algorithm to optimize the entire multi-stage query pipeline within a single platform

Data Source

Leveraging arXiv's comprehensive academic repository for hybrid search implementation.

Dataset

arXiv open-access archive for scholarly articles

Scale

2.7M+
PAPERS

130
DISCIPLINES

Spanning Computer Science, Physics, Mathematics, etc.

https://www.kaggle.com/datasets/Cornell-University/arxiv

EXPLOITED DATA STRUCTURE

Title & Abstract

Stored as **plain text** and **dense vectors** to serve different search algorithms.

Metadata

Publication Year and **Category** fields used for building strict filters.

Problems & Elasticsearch Solutions

Addressing search limitations with advanced hybrid architecture capabilities.

PROBLEM 1: THE RIGIDITY OF KEYWORD SEARCH

Users must type exact matches; synonyms or rephrasing return zero results.

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ELASTICSEARCH SOLUTION

Provides **Vector Search (kNN)** capabilities to understand the 'contextual meaning' of the query rather than just matching word for word.

PROBLEM 2: AI MISSES SPECIALIZED TERMINOLOGY

AI understands context well but is poor at finding exact author names or rare acronyms.

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ELASTICSEARCH SOLUTION

Runs both streams in parallel **(Hybrid Search)**. It natively integrates the **Reciprocal Rank Fusion (RRF)** algorithm to automatically score and merge AI and keyword results with high precision.

PROBLEM 3: SEARCH SKEWED BY MULTIDISCIPLINARY NOISE

A single keyword can have multiple meanings. Applying strict filters (year, category) to AI systems often breaks the search structure, causing missing results or high latency.

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ELASTICSEARCH SOLUTION

Features an **Automatic Optimization filtering** mechanism. It dynamically switches between scanning algorithms based on filter strictness, ensuring fast and accurate responses.